# Auditing World-Model Explanations in Retail Forecasting: Faithfulness, Accuracy and Causal Validity

Karthik Iyer

Correspondence: karthik.iyer.au@gmail.com ORCID: 0009-0004-0593-1602

**Abstract.** Retail forecasting models that see the price and promotion plan can explain each forecast as a baseline, what would sell with nothing planned, plus the lift each part of the plan adds. Merchants read that split as fact and plan on it. We set out eight checks on such explanations, fixed in advance: properties that hold by construction, sensitivity to randomised weights, stability across seeds, price direction, agreement with what sold, agreement with LightGBM, a plan read-back, and agreement with a measured cause. In generated store data with known effects, a world model whose explanation is built in passes C1 and C3 to C6, and its lifts match the truth (rank agreement 0.99) when plans are made at random. When merchants plan into high-demand weeks, it still passes them, forecasts more accurately, and overstates the total lift by a third (1.33 to 1.40 times) and the display's by 1.81 to 1.93 times; LightGBM overstates them too, and the two agree. Asked about a lasting change in the regular price, the world model answers with the temporary-cut response, $-2.42$ to $-2.37$ against a true $-0.5$, and only the measured cause finds it. The price-direction check catches every model blind to price but also flags LightGBM; an exploratory lever-by-lever comparison catches lifts given to the wrong lever; the randomisation check separates nothing. On Dominick's we find 54 spells in which a store's regular prices sat apart from its zone peers, most between April 1993 and April 1994. Over the 38 scored spells, prices 6% to 10% off their peers, the pooled effect on category units is $-0.33$ over eight weeks (95% interval $-0.68$ to $0.01$) and $-0.52$ over the full spell ($-0.81$ to $-0.23$), in line with the elasticity near $-0.3$ the chain's experiments implied. This is a noisy observational benchmark, not a ground truth: the treated stores are inferred, prices move before the detected start, and a placebo at false start dates gives $-0.26$ ($-0.72$ to $0.19$), close to the 8-week estimate. We then audited all 494 saved world-model and lift-model versions of our companion paper on M5, Dominick's and Breakfast at the Frat. No version passed every check on M5 or Dominick's. The baseline and lift add up to the forecast in every version, and under the model's own plan for nothing planned the lift is zero; under the audit's plan it was not, on most Dominick's and Frat versions, and every such failure sits where the two plans differ: some configurations take the regular price from a held or recorded price, and for items with no recent price the models compute the baseline at the item's mean price or at 1.0. On Dominick's and M5 a deeper price cut lowered the forecast in about half the item-weeks or more; and on M5 the lifts agreed poorly between seeds although the forecasts agreed. Both models state about twice the measured response to the zone price deviations, though the measured interval is wide and overlaps both. Comparing configurations on the same data and window, with the dependence between related configurations kept, the number of checks passed was not clearly related to accuracy (pooled Spearman $-0.10$, 95% interval $-0.21$ to $0.02$); with C5, which compares the forecast with sales, left out, $-0.02$ ($-0.13$ to $0.12$). These results hold for these models and data sets.

**Keywords:** retail forecasting; explanation; faithfulness; promotions; price response; difference in differences; world models.

## 1 Introduction

A retail forecast is used for more than its number. When a merchant plans a promotion, the question is not only how much will sell but how much of it the promotion will add, and whether a deeper price cut would add more. Forecasting models that see the plan can answer that directly: they forecast the item with the plan as made, and again with nothing planned, and the difference is the lift the plan adds. The world model and the lift model of our companion paper [34] are built this way: each forecast is a baseline under the nothing-planned plan plus a lift, and the lift is exactly zero when nothing is planned. The split is what a merchant reads ("the promotion added 40 units"), plans on, and is judged by.

Such an explanation is useful only if it can be trusted, and being built into the model does not make it right. The explanation literature separates two properties [1, 36]: an explanation is *faithful* if it reflects what the model does, and *correct* if it reflects what happens in the world. In molecular property prediction, MolSanity found the two come apart: attributions faithful to the model were often anti-aligned with the substructure that determines the property [2]. Retail has a property chemistry lacks. Some causes can be measured: Dominick's Finer Foods moved store prices on purpose in the early 1990s [3], so

the response of sales to a price change can be estimated from the data and a model's stated response compared with it.

This paper audits plan-lift explanations in retail forecasting. It asks three questions, fixed in a plan before any result [35]: are the explanations faithful to the model, stable and sensible; do they match what sold and what a price change causes; and across model versions, do versions whose explanations pass more checks also forecast better?

**Contributions.**

1. A definition of the plan-lift explanation of a forecast that applies to any model that reads the plan, and eight checks on it (Figure 1), each with a pass rule fixed in advance.
2. A known-answer study: generated store data in which the effect of every part of the plan is known, with failures planted on purpose, to measure which checks catch which failure before they are trusted on real data.
3. A noisy observational benchmark on public retail data: 54 spells in six Dominick's categories where one store's regular prices sat apart from its zone peers, 38 of them scored, with their effect on sales measured by difference in differences, an event study, a placebo and a sensitivity analysis. Which stores were treated is inferred, prices move before the detected start, and a placebo at false dates gives a slope close to the estimate, so it is not a ground truth; the causal failure is established in the known-answer study.
4. An audit of every saved world-model and lift-model version of the companion paper on M5, Dominick's and Frat, against LightGBM, and the test of whether faithful explanations go with accurate forecasts.

**Findings.** In generated data with known effects, explanations that pass every check against the model, the data and another model can still be wrong in size, when merchants plan into strong weeks, and wrong in kind, when a model answers a question about the regular price with what it learned from temporary cuts. The cheap price-direction check and a lever-by-lever comparison with a second model catch the cruder failures; the randomisation check does not test an explanation built into the model; and only a measured cause finds the error in kind (§4). On Dominick's, zone price deviations give a pooled 8-week response close to a placebo at false dates, a larger full-spell response whose interval excludes zero, and single episodes that are noise: a noisy benchmark, not a ground truth (§6). In the audit of the companion paper's 494 saved versions (§7), no version passed every check on M5 or Dominick's; on those data sets a deeper price cut lowered the forecast in about half the item-weeks or more; both the world model and LightGBM stated about twice the measured response to a regular price change, though the measured interval is wide; every failure of the lift to be zero with nothing planned sits where the audit's plan for nothing planned differs from the model's own, including items whose baseline the models compute at a price no merchant would call regular; and, among configurations compared on the same data and window, the number of checks passed was not clearly related to accuracy; the intervals exclude a strong relation, not a weak one. These are findings for these models on these public data sets.

## 2 Related work

**What an explanation should satisfy.** Interpretability lacks a single definition [38, 39]. Jacovi and Goldberg [1] separate faithfulness, whether an explanation reflects the model's reasoning, from plausibility, whether it convinces a person, and warn that evaluations often measure the second. Chen et al. [36] draw the related line between explanations true to the model and true to the data, and Janzing et al. [37] argue that feature relevance is a causal question about interventions on the model's inputs. Our on/off contributions are interventions of exactly that kind, made on the plan inputs the merchant controls.

**Feature attributions and their failures.** LIME [4] and SHAP [5], with its exact form for trees [6], are the most used attributions [7]. Saliency maps can barely change when a network's weights are randomised [8], move with input shifts the model ignores [9], differ for near-identical inputs [10], and rank features whose removal does not hurt the model [11]. Rudin [12] argues for models that are interpretable by construction. The world model and the lift model are of that kind for the plan, which lets this audit compare a built-in explanation with post-hoc ones on the same forecasts.

**Explanations of time series and forecasts.** Rojat et al. [13] survey explanation methods for time series. Ismail et al. [14] and Tonekaboni et al. [15] show on generated data with known important time steps that saliency methods designed for images often fail on temporal data, the approach our known-answer study takes for retail plans. Among forecasting models, the Temporal Fusion Transformer [26] reports attention weights as explanations. We know of no study that checks the

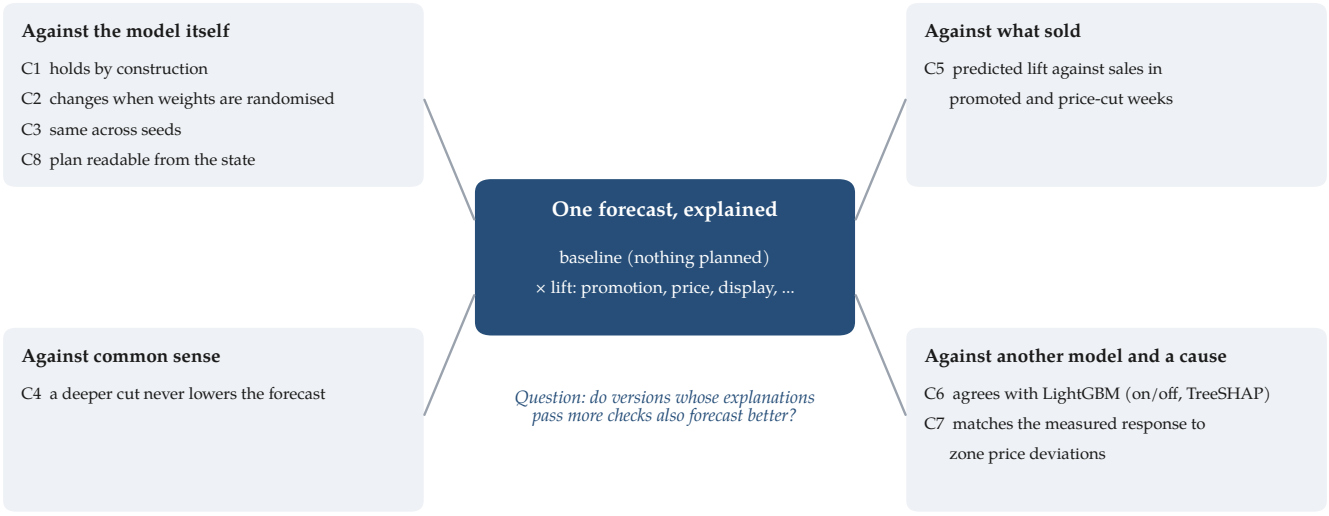



**Figure 1: Each forecast's explanation is checked against the model itself, common sense, what sold, another model and a measured cause.** The eight checks are fixed in `docs/PAPER2_PLAN.md` before any result; every model version and every check is reported.

explanation of a demand forecast against a measured promotional or price effect.

**Promotion and price effects.** Marketing science has measured promotional effects for decades: the split of a promotion's sales into brand switching, timing and quantity [32, 33], the dip after a promotion [30, 29], and the size of price elasticities and what drives it [40]. Promotion forecasting systems built on these effects include PromoCast [28]; Ma, Fildes and Huang [31] and Fildes et al. [27] review the forecasting side. These elasticities describe temporary cuts. Regular price changes move far fewer units: in Dominick's randomised store-level pricing experiments [3], regular prices were raised or lowered 10% across many categories, and volume moved about 3% over the following 16 weeks, an elasticity near $-0.3$. Hoch et al. [21] and Montgomery [16] model store-level price response on the same data.

**Quasi-experiments in scanner data.** The public Dominick's files do not mark the experimental stores or weeks. Abrams, Gui and Hortaçsu [17] reconstructed them with the original documentation and found that some published price changes did not replicate. We therefore treat the price deviations visible in the public data as a quasi-experiment, never as the experiments, and estimate their effect by difference in differences [18] with standard errors clustered by store [19, 20].

## 3 The explanation and the checks

### 3.1 Notation

A data set has series $i$ (an item in a store) observed in periods $t$. The plan for series $i$ in period $t$ is $x_{it} = (p_{it}, \ell^1_{it}, \dots, \ell^L_{it})$: the shelf price and $L$ lever indicators (a deal flag on Dominick's; promotion, display, feature and a tag-only price reduction flag on Frat; event and SNAP days on M5). A model fitted on the periods before an origin forecasts the next $H$ periods from the history and the plan for those periods, $\hat{y}_{it} = f_{it}(x)$.

The *regular price* $r_i$ is the series' highest price over the last 12 training periods, the rule the models themselves use. The plan with *nothing planned*, $x^0$, sets every price to $r_i$ and every lever to zero for all $H$ periods. The explanation of a forecast has four parts:

$$b_{it} = f_{it}(x^0) \qquad \text{baseline} \tag{1}$$

$$\Lambda_{it} = f_{it}(x) - b_{it} \qquad \text{total lift} \tag{2}$$

$$c^k_{it} = f_{it}(x) - f_{it}(x \mid \ell^k = 0) \qquad \text{contribution of lever } k \tag{3}$$

$$c^p_{it} = f_{it}(x) - f_{it}(x \mid p = r) \qquad \text{contribution of the price} \tag{4}$$

and the interaction $\iota_{it} = \Lambda_{it} - c^p_{it} - \sum_k c^k_{it}$, which is zero only when the levers act separately. Switching a lever off switches it off for all $H$ periods, so the dip after a promotion is part of that promotion's contribution. For the world model and the lift model the forecast is built as $\hat{y} = B + \lambda$, a baseline readout and a lift readout [34]; for LightGBM [25] the same on/off quantities are computed from its forecasts, together with TreeSHAP values $\phi^k_{it}$ [6] on its lever and price inputs. Unlike Shapley values, on/off contributions answer the merchant's ques-

tion (what would sell without this lever) directly, and are defined for any model.

Rank agreement $\rho(a, b)$ below is the Spearman correlation over item-weeks where either $a$ or $b$ is non-zero, so that the many weeks with nothing planned do not inflate it.

### 3.2 The eight checks

**C1 By construction..** With $\hat{y} = B + \lambda$, three violations, each relative to the largest forecast: the lift with nothing planned, $\max|\lambda(x^0)|$; the movement of the baseline with the plan, $\max|B(x) - B(x^0)|$; and $\max|B + \lambda - \hat{y}|$. These hold by design and are checked because a design can be broken by an implementation.

**C2 Randomisation..** Following Adebayo et al. [8], the network's parts are re-initialised in turn from the output back to the input: lift readout, baseline readout, transition, encoder. After each step, the rank agreement of the total lift with the original, $\rho(\Lambda, \Lambda^{(k)})$, should fall. An explanation that survives randomisation does not depend on what the model learned.

**C3 Seed stability..** Fits of the same configuration that differ only in their seed should explain a forecast alike. The score is the mean over seed pairs of $\rho(\Lambda^s, \Lambda^{s'})$, shown against the same agreement for the forecasts themselves.

**C4 Price direction..** For depths $0 < 5\% < 10\% < 20\% < 30\%$ applied to the planned price, the share of item-weeks and steps in which the deeper cut gives a lower forecast than the shallower one by more than 0.1% of the forecast.

**C5 Against what sold..** On item-weeks with a lever on or a price at least 5% below regular, the bias $\sum \hat{y} / \sum y$ and the rank agreement of predicted lift $\hat{y}/b$ with observed lift $y/b$. Merchants plan promotions for weeks they expect to sell well, so $y/b$ overstates what the plan caused; this check measures agreement with the data, not with a cause.

**C6 Against LightGBM..** For each lever, $\rho(c^k, c^k_{\text{LGB}})$ and $\rho(c^k, \phi^k)$, on the same window. Two models fitted to the same data can agree and both be wrong; agreement is evidence of a shared signal, not of correctness.

**C7 Against a measured cause..** For a spell $e$ in which store $s$'s regular prices sat at a log ratio $\delta_e$ from its zone peers, the model's stated response is

$$\hat{R}_e = \log \frac{\sum_{t \in E} \sum_{i \in s} f_{it}(x)}{\sum_{t \in E} \sum_{i \in s} f_{it}(x \mid p_s = p_s e^{-\delta_e})}, \qquad (5)$$

the change in the store's forecast category units if its prices had stayed with its zone, over the spell's first eight weeks $E$. It is compared with the measured response $\tau_e$ from difference in differences (§6): whether $\hat{R}_e$ falls inside $\tau_e$'s 95% interval, whether its sign agrees, and the pooled slope of each on $\delta_e$.

**C8 Plan read-back..** For the world model only (paper 1): the $R^2$ of a ridge probe that reads the day's discount back from the frozen model's state after it has rolled through the plan, against the same probe on the state before the window, which has not seen the plan.

**Table 1: Every check has a pass rule fixed before any result.** Scores as defined in §3; a check that does not apply to a model (C1 and C2 for LightGBM) is not counted for it.

| Check | Passes when | Applies to |
|---|---|---|
| C1 | every violation $< 0.001$ | lift readout models |
| C2 | $\rho$ after the lift readout $< 0.5$ | neural models |
| C3 | mean seed $\rho \geq 0.7$ | configurations with $\geq 2$ seeds |
| C4 | share $< 1\%$ | all |
| C5 | bias in $[0.8, 1.25]$ and rank $\geq 0.3$ | all |
| C6 | $\rho(\Lambda, \Lambda_{\text{LGB}}) \geq 0.5$ | non-LightGBM |

### 3.3 Pass thresholds and the main test

Thresholds were fixed in the plan before any audit result was read (Table 1). For the main question, each model version (a configuration, seed and window) gets the number of checks it passes among C1 to C6, counting only those that apply to it. Within each data set the Spearman correlation across versions between that count and item-by-store WRMSSE (lower is better) is computed, and the same for each check's continuous score. A negative correlation means versions that pass more checks forecast better. The data sets are pooled by a permutation test that shuffles the scores within each data set, 10,000 times. C7 and C8 apply to a few models each and are reported per model.

## 4 A known-answer study

Before the checks are trusted on real data, they are run where the answer is known. Following Ismail et al. [14] and Tonekaboni et al. [15], who test time-series explanations on generated data with known important inputs, we generate weekly store data in which the effect of every part of the plan is set by us, fit correct models and models with failures planted on purpose, and ask two questions: which checks catch which failure, and do the explanations of models that pass the checks match the truth?

### 4.1 Design

**The data.** Three price zones of six stores sell 20 items for 260 weeks (Figure 2). Expected sales are the item's size and season times an unobserved demand shock (a first-order autoregression on the log scale, standard deviation 0.25, persistence 0.7), times the effects in Table 2. A temporary cut and a change in the regular price have different elasticities, $-2.5$ and $-0.5$, as in the literature on promotions and the Dominick's experiments. Each item gets one chain-wide regular price rise of 3% to 8%. In each zone one store's regular prices rise 10% and one store's fall 10% for 16 weeks at random times during training, and one rises and one falls starting at the forecast origin, week 253, so that every model can be scored on them exactly as on Dominick's. Sales are Poisson around their expectation.

**Two worlds.** In the first, cuts, promotions and displays are planned at random. In the second, merchants plan them more often in weeks when the unobserved demand shock is high (the log odds of each rise by the shock in standard deviations), as merchants do when they promote into strong weeks. Everything else is the same.

**The models.** In each world: the world model with three seeds; the world model fitted with each series' prices shuffled across its training weeks, so that it cannot learn a price effect (*blind to price*); the world model fitted with promotion and display shuffled the same way (*blind to levers*); LightGBM; and LightGBM blind to price. The forecast for weeks 253 to 260 is explained and every check run as on real data; the world model has the settings of the companion paper's Dominick's world model except that it is a single fit rather than an average of three, without store switching, with width 32 and 2,000 steps, to fit on a processor; LightGBM has the companion paper's Dominick's settings. Each explanation is also scored against the truth, computed from the expected sales under the same counterfactual plans: rank agreement and the ratio of summed lifts on the 1,743 of 2,880 item-weeks with something planned, overall and by lever.

### 4.2 What the checks catch

Figure 3 shows every check on every model, and Table 7 in the appendix every score. Six results stand out.

**Correct models pass, and their explanations are right.** With plans at random, the three world-model seeds pass every check that applies except C7 and, for one seed, C2, and their lifts match the truth: rank agreement 0.99 and summed lift 1.14 times the true lift for seed 1, with the price, promotion and display parts each within 13% of the truth. Seeds agree on their lifts as closely as on their forecasts (C3, 0.99). LightGBM's lifts rank 0.85 with the truth at 1.09 times.

**Table 2: The generated data have known effects for every part of the plan.**

| Effect on expected sales | Size |
|---|---|
| Temporary cut, elasticity on price/regular | $-2.5$ |
| Regular price, elasticity on regular/list | $-0.5$ |
| Promotion | $+40\%$ |
| Display | $+25\%$ |
| Promotion and display together, extra | $+15\%$ |
| Week after a planned week with nothing planned | $-15\%$ |
| Chance of a cut; depth | 12%; 10% to 40% |
| Promotion chance, with a cut; without | 60%; 3% |
| Chance of a display | 8% |

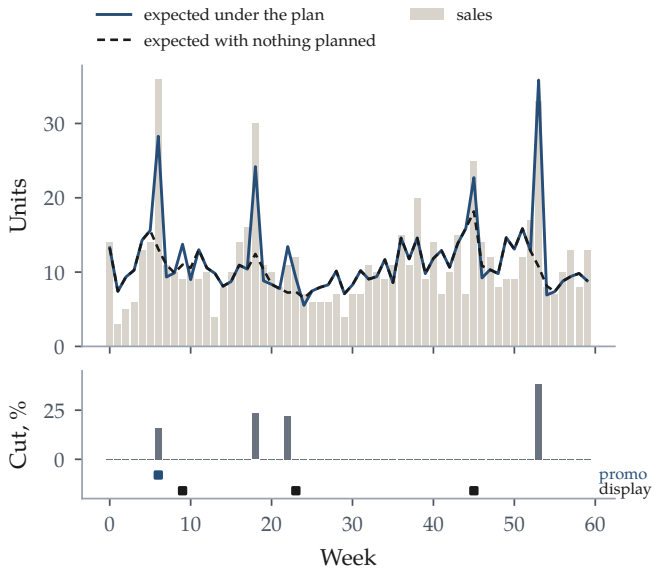



**Figure 2: In the generated data the effect of every part of the plan is known.** One item in one store over 60 weeks, in the world where merchants plan into high-demand weeks: sales, expected sales under the plan, and expected sales with nothing planned; below, the cut and the weeks with a promotion or a display.

**The price check catches a model blind to price, and flags LightGBM too; the check against sales does not.** All four models fitted without a usable price fail C4: 31.5% and 27.2% of item-weeks for the world model, 49.5% and 43.4% for LightGBM, against 0% for the world models fitted on real prices. It also fails both LightGBM models fitted on real prices (7.6% and 9.9%), whose lifts are close to the truth and whose regular-price response passes C7: trees respond to price in steps, so a deeper cut sometimes lowers the forecast. C4 is sensitive to a

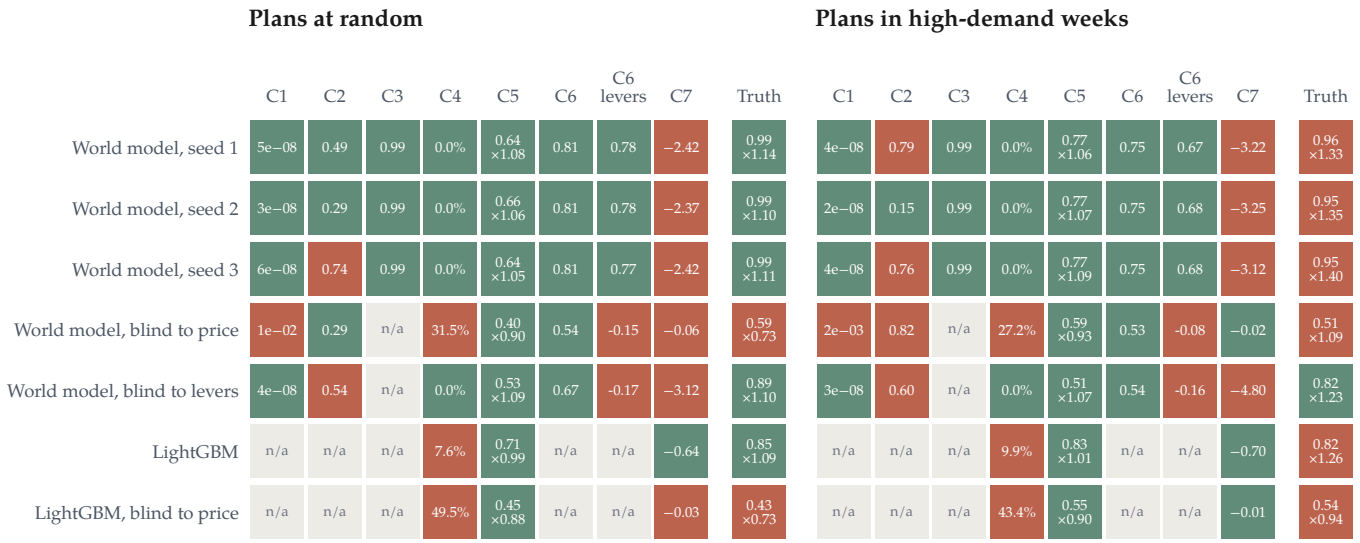



**Figure 3: Correct models pass the checks; so do models whose explanations are wrong in ways only some checks can see.** Each check's score and its pass (green) or fail (red) under the rules fixed in advance, for every model in both worlds; grey, does not apply. C5: rank of predicted with observed lift, and summed forecast over summed sales. C6 levers: the lowest per-lever rank agreement with LightGBM, reported but without a rule fixed in advance. C7: pooled slope of the stated response to the price spells at the origin, passed when inside the 95% interval of the measured slope. Truth: rank agreement of the model's lifts with the true lifts and the ratio of their sums, green when the rank is at least 0.7 and the ratio within 0.8 to 1.25.

model ignoring price but not specific to it. All four pass C5: their forecasts on planned weeks sum to 0.88 to 0.93 of sales, with rank agreement 0.40 to 0.59. Their explanations are wrong in a way a merchant would act on: the price part is zero, and the promotion, which usually comes with a cut, takes its place, at 1.95 to 2.37 times its true lift across the four (Figure 4). Both price-blind world models also fail C1 narrowly, because the audit takes the regular price from the true history while these models learned it from a shuffled one; that is a property of how the failure was planted, not a sign that C1 detects it.

**Only a per-lever comparison catches lifts given to the wrong lever.** The per-lever comparison was added after the first results, without a pass rule fixed in advance, and is exploratory. The world model fitted without usable promotion and display flags passes C1, C4 and C5, and its total lift agrees with LightGBM's (C6, 0.67 and 0.54) and ranks 0.89 with the truth. Its explanation gives the promotion 0.14 and the display 0.05 times their true lift, and the price 1.35 times. Every check on the total lift except C2 passes it. Its per-lever agreement with LightGBM does not: 0.29 for the promotion and −0.17 for the display, against 0.80 for the correct world model.

**When plans follow demand, every check passes and the lifts are a third too large.** In the second world the three world-model seeds pass C1, C3, C4, C5 and C6, and their forecasts are more accurate than in the first (item-by-store RMSSE 0.345 against 0.401): a plan made in high-demand weeks tells the model about the demand. Their lifts are 1.33 to 1.40 times the true lift, and the display's 1.81 to 1.93 times. C5 cannot see this: sales in planned weeks are high for the same reason, and the forecasts sum to 1.06 times them. C6 cannot see it either: LightGBM, fitted to the same data, overstates by 1.26 times, and the two agree at 0.75. No check that compares the explanation with the model, with the data or with another model fitted to the same data can find a bias that the data themselves carry.

**The world model treats a regular price change like a temporary cut.** For the price spells that start at the origin, the world model's stated response is $-2.42$ to $-2.37$ per unit of log price across seeds with plans at random and $-3.25$ to $-3.12$ when plans follow demand, 4.7 to 6.5 times the true $-0.5$; and $-2.5$ is the temporary-cut elasticity (Figure 5). It has learned how sales answer a cut and applies it to a change in the regular price. LightGBM, whose inputs include the item's recent prices, states $-0.64$ and $-0.70$. The measured response from difference in differences on the generated sales is $-0.73$ ($-1.41$ to $-0.06$) and $-0.74$ ($-1.90$ to $0.41$): imprecise with six spells and three or four controls each, as on Dominick's, but enough to reject the world model's answer in both worlds and not LightGBM's. In the second world the measured interval also contains zero, so there C7 passes the price-blind models too: with six spells it can reject a response five times too large, not one that is absent. Apart from C2 for one seed, C7 is the only check the world model fails in the first world, and the only one

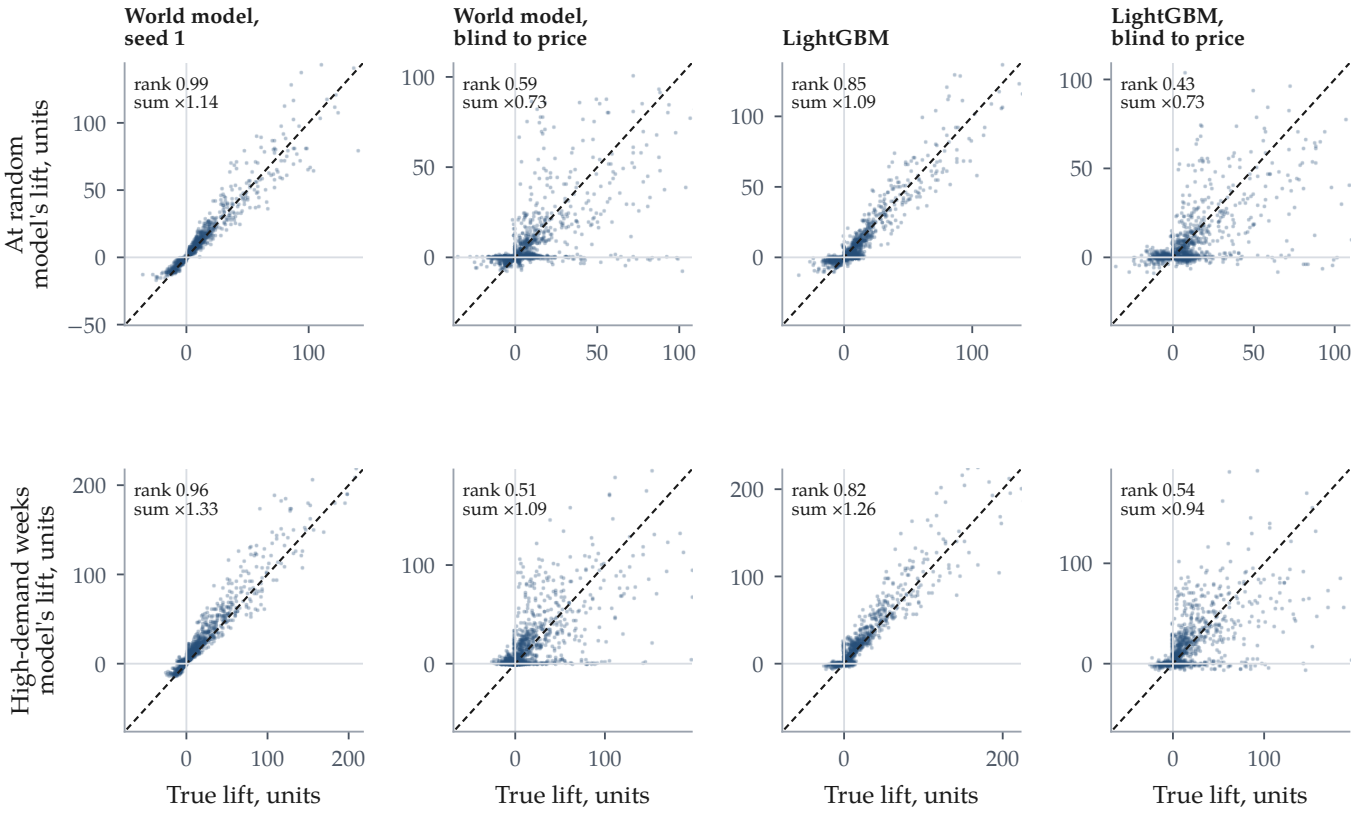



**Figure 4: When plans follow demand, both the world model and LightGBM overstate what the plan adds, and agree with each other.** Each point is an item-week with something planned: the model's total lift against the true lift, units. Dashed: equal. Top, plans at random; bottom, plans made more often in high-demand weeks.

that finds this error.

**Randomisation does not separate working models from broken ones.** After the lift readout is re-initialised, the rank agreement of the lifts with the original is 0.15 to 0.82 across world models, and with every part re-initialised it is 0.44 to 0.70 (Figure 6). The construction itself ties the lifts to the weeks with something planned and to the size of the item, so a network with random weights still ranks them much as the trained one does. C2 fails correct seeds and passes broken models; as specified, it does not test a lift built into the model, and we do not use it as evidence below.

**What the study says about the checks.** Across the 14 models, the share of checks passed tracks how well a model ranks the true lifts (Spearman 0.87) but not how close its lifts are in size ($-0.27$ with the negative absolute log ratio): the models that pass the most checks include the confounded ones whose lifts are a third too large. Accuracy tracks the gross failures, the price-blind models being the least accurate, and misses the subtle ones. The checks earn their place as follows. C4 finds a model that ignores price, and also flags one whose price response is merely uneven. Per-lever agreement with another model finds lifts given to the wrong lever. C1 and C3 confirm construction and stability. C5 and C6 confirm agreement but cannot find a bias the data carry, and C2 as specified finds nothing. Only a measured cause, C7, finds that a model's stated response to a price change is wrong, and only one kind of response, a sustained regular-price change, has a measured cause on public retail data.

## 5 Data, model versions and protocol

**Data.** The three public data sets of the companion paper [34]: M5, daily Walmart sales of 3,049 items in 10 stores from 2011 to 2016, with prices, event days and SNAP days [24]; Dominick's Finer Foods, weekly scanner data by store for a Chicago chain from 1989 to 1997, with price, a promotion code and cost [22], of which the companion paper's six final categories are audited (cheese, snack crackers, soft drinks, bottled juice, analgesics, frozen entrees); and Breakfast at the Frat, weekly sales of 55 breakfast items in 77 stores from 2009 to 2012, with display, feature and tag-only price reduction flags [23]. No raw data is redistributed.

**Model versions.** Every saved world-model and lift-model checkpoint from the companion paper is audited:

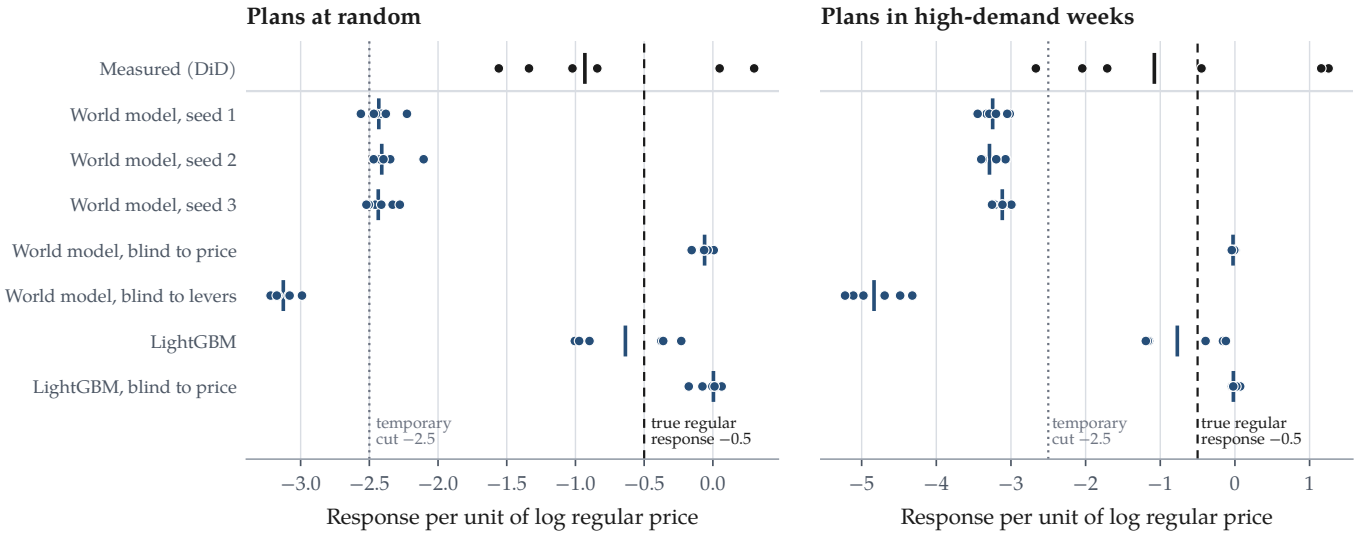



**Figure 5: Every world model states the response to a regular price change as if it were a temporary cut; LightGBM does not.** The stated change in store category units per unit of log price for each of the six price spells that start at the forecast origin (dots; bar, median), against the measured response from difference in differences on the generated sales (top row), the true response ($-0.5$, dashed) and the temporary-cut elasticity ($-2.5$, dotted).

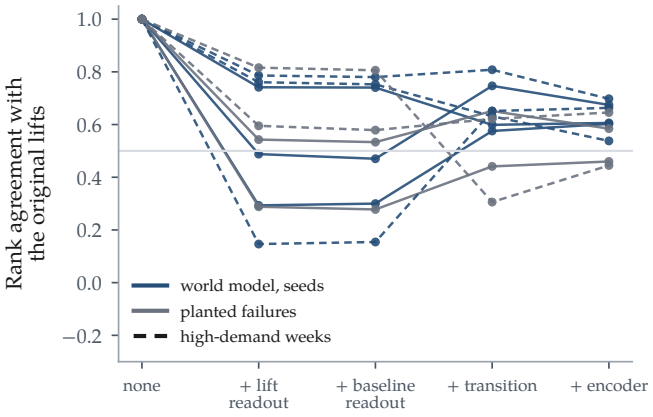



**Figure 6: Re-initialising the network leaves the lifts' ranking largely in place.** C2 for every world model, both worlds: rank agreement of the total lift with the original after the lift readout, then the baseline readout, the transition and the encoder are re-initialised in turn.

the development and final windows of Frat, the development versions on Dominick's, and the M5 practice, confirmation and test windows. The Dominick's final world models, each an average of three fits, were not saved in the companion paper; they are refitted with settings unchanged for the audit, three seeds each, and the companion paper keeps the scores of its original runs. Each version's accuracy is read from its own run record. LightGBM is refitted on the Dominick's and Frat final windows and the M5 test month for C6, with the settings of the companion paper.

**Protocol.** The audit reads models and never tunes them. Every check runs on every version it applies to and every result is reported, including failures. The plan, its thresholds and every amendment are dated in `docs/PAPER2_PLAN.md`; each amendment was made before the results it affects were read.

## 6 An observational benchmark from zone price deviations

### 6.1 Finding the deviations

Dominick's set regular prices by zone: within a zone every store was meant to carry the same regular price [22]. For each store and item we take the log of the week's regular price (no deal code, not more than 5% below the item's highest price in that store over the last 12 weeks) over that item's median regular price in that store over all weeks, and subtract the median of the same across the zone's stores that week. Moves made zone-wide cancel, and so do an item's usual price differences between stores; a store moved on its own stands out. An episode is at least 12 consecutive weeks in which this quantity is at least 3% in the same direction for at least 60% of the store's items, with a median of at least 5% over the episode. Controls are stores of the same zone within 1% of their zone over those weeks. An episode with fewer than three controls, or without the 26 comparison weeks before its start, is listed but not scored.

The detector was written, and its episodes listed, before any model was scored against them. A first version measured stores against the zone's most common price and found one episode, in analgesics: in the experiments a third of a zone's stores moved up and a third down, so the most common price was often a moved one. The

change is recorded, with its date, in the plan.

### 6.2 Measuring their effect

For episode $e$ of store $s$ with controls $\mathcal{C}_e$, the response is the difference in differences on log category units,

$$\log Y_{jw} = \alpha_j + \gamma_w + \tau_e D_{jw} + \varepsilon_{jw}, \qquad j \in \{s\} \cup \mathcal{C}_e, \quad (6)$$

over the 26 weeks before the episode and its first eight weeks, where $Y_{jw}$ is store $j$'s category units in week $w$ and $D_{jw}$ marks the treated store in the episode weeks. Standard errors are clustered by store [19, 20] with the CR1 correction and $t$ critical values on stores minus one degrees of freedom; with 4 to 25 stores per episode, single intervals are wide by construction. The plan scores each episode by its own interval; after the placebo showed those intervals to be too narrow, the pooled slope of $\tau_e$ on $\delta_e$ across episodes, with an interval robust to unequal variances, was added as the summary used for C7. The estimate over the full episode is kept in the list as well. The models forecast at most eight weeks ahead, so each model is fitted on the data up to the week before an episode and scored on its first eight weeks, the same weeks as $\tau_e$.

### 6.3 What the deviations show

The detector finds 54 episodes in the six categories, 38 with at least three control stores (Figure 7). Most lie between Dominick's weeks 190 and 240, April 1993 to April 1994. Hoch, Drèze and Purk published their experiments in 1994, and the public files do not date them, so whether these spells are those experiments cannot be checked. The scored stores' prices sit 6% to 10% off their zone peers. Over the first eight weeks of each scored episode, category units move against price in 63% of episodes. The pooled slope of the measured response on the price change is $-0.33$, with a 95% interval of $-0.68$ to $0.01$ (heteroskedasticity-robust across episodes), and over the full episode $-0.52$ ($-0.81$ to $-0.23$). Both are in line with the elasticity near $-0.3$ implied by Hoch, Drèze and Purk's report of a 3% volume change for a 10% price change, and neither is precise. These intervals across episodes were added after the placebo was seen (below).

Three robustness analyses, fixed in advance, and two added after seeing them qualify this benchmark (Figure 8).

**Prices move before the detected start.** The detector dates an episode from the first week in which most of the store's items are at least 3% off. Prices ease in before that: in the eight weeks before the detected start, 47% of store-weeks are already 3% or more off in the episode's direction (panel a). Part of the 26 comparison weeks is therefore treated, which biases the 8-week estimate towards zero and shows in the event study as a gradual decline in signed sales over the weeks before the start (panel b). The full-episode estimate, which compares the whole spell, is larger, $-0.52$ against $-0.33$.

**Table 3: The pooled response keeps its sign under every detector setting.** Episodes found and scored, and the pooled 8-week slope, for each weekly size and episode median; the plan's setting in bold.

| Each week | Episode median | Episodes | Scored | Slope |
|---|---|---|---|---|
| 2% | 4% | 64 | 45 | −0.38 |
| 2% | 5% | 55 | 38 | −0.37 |
| 2% | 7% | 38 | 26 | −0.20 |
| 3% | 4% | 63 | 45 | −0.32 |
| **3%** | **5%** | **54** | **38** | −0.33 |
| 3% | 7% | 37 | 26 | −0.16 |
| 4% | 4% | 59 | 43 | −0.32 |
| 4% | 5% | 54 | 38 | −0.34 |
| 4% | 7% | 37 | 26 | −0.17 |

**Single episodes are noise.** Moved 26 weeks earlier, to weeks where prices had not moved, the same estimator gives a pooled slope of $-0.26$ ($-0.72$ to $0.19$), and 42% of the 38 placebo intervals exclude zero, where 5% should. The intervals clustered by store are far too narrow: their median standard error is 0.02 while placebo estimates spread with a standard deviation of 0.12. With few stores per episode, weekly category sales that move together over many weeks, and errors clustered by so few stores, this is the failure Bertrand, Duflo and Mullainathan [19] and Cameron and Miller [20] describe. A single episode's interval, and with it the plan's per-episode score for C7, cannot be trusted; the comparison of models with the measured response is made on the pooled slope and its interval.

**The pooled response is stable but imprecise.** Over the nine detector settings (Table 3) the number of scored episodes ranges from 26 to 45 and the pooled slope from $-0.38$ to $-0.16$. By category, the slope is negative in 3 of the 3 categories with at least five episodes, and every category's interval is wide (panel d).

What survives is a pooled response of store category units to a sustained change in regular prices of $-0.33$ over the first eight weeks, whose interval just includes zero and overlaps the placebo's, and $-0.52$ over the full spell, whose interval excludes zero. The 8-week figure is pulled towards zero by prices that moved before the detected start. That is enough to tell a model that treats a regular price change like a temporary cut, an elastic-

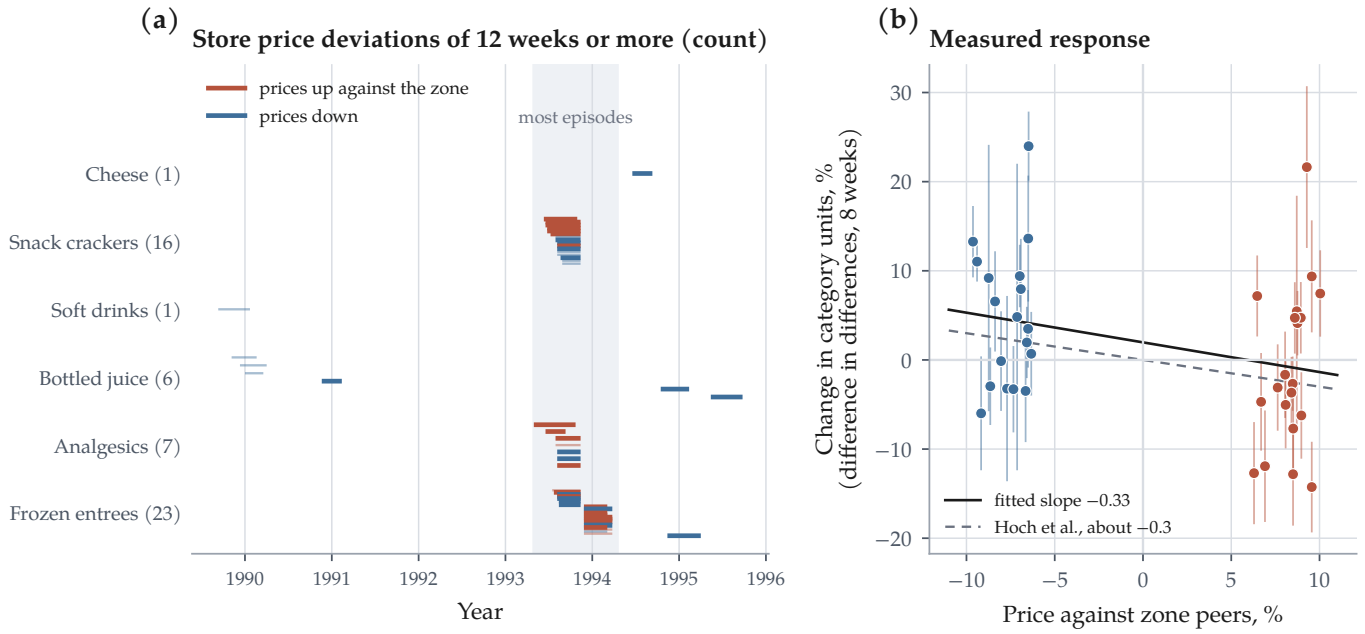



**Figure 7: The price deviations cluster between spring 1993 and spring 1994, and on average sales move against price.** (**a**) Every episode found in the six categories (count in brackets); faint: fewer than three control stores, listed but not scored. (**b**) For the 38 scored episodes, the store's price against its zone peers and the measured change in its category units over the episode's first 8 weeks (difference in differences against same-zone stores, store and week fixed effects, 95% interval clustered by store; these intervals are too narrow, see Figure 8). Solid line: fitted slope; dashed: the elasticity near $-0.3$ implied by the experiments [3].

ity several times larger, from one that does not; it is not enough to tell a model with no price response from a correct one. That is the comparison C7 can make on public data.

**What this benchmark is, and is not.** It is not a ground truth. Which stores and weeks were treated is inferred from prices, not recorded; prices move before the detected start; and the placebo at false start dates gives a pooled slope ($-0.26$) close to the 8-week estimate ($-0.33$), so the 8-week estimate cannot be told from what the estimator finds where nothing happened. The full-spell estimate ($-0.52$, $-0.81$ to $-0.23$) has an interval that excludes zero, but it too overlaps the placebo's ($-0.72$ to $0.19$). These deviations may include zone moves that were not experiments, and the models were trained on data that include earlier deviations. What the benchmark measures, noisily, is the response of a store's category units to a sustained change in its regular prices relative to its zone, over eight weeks. It is not the response to a temporary cut, which the companion paper's models mostly learn from, and a model can be right about one and wrong about the other. The known-answer study (§4) has both, with different true elasticities, for that reason.

## 7 The audit of the saved model versions

The audit read every saved world-model and lift-model checkpoint of the companion paper [34]: 494 versions (a configuration and seed), 31 on M5, 279 on Dominick's and 184 on Frat, scored on 933 version-windows. Most are development variants, so they differ in many settings at once. No model was refitted or tuned. Table 4 gives the pass counts and Figure 9 every score.

**No version passes every check on M5 or Dominick's.** On Frat 35 of 362 version-windows pass every check that applies; on M5 and Dominick's none do. The checks fail for different reasons on each data set, and two of them, C1 and C4, needed a diagnosis before they could be read. C4 was rerun on 14 saved runs (24 windows) chosen by a rule fixed before the rerun (`docs/PAPER2_PLAN.md`, 17 October), and C1 was traced for every audited configuration (18 October); both diagnoses are reported as added after the results.

**C1: the parts add up, and every failure traces to what "nothing planned" means.** In every version baseline plus lift equals the forecast (largest violation 2e-16 of the largest forecast), and the baseline moves with the plan by at most 1.7% of it. The lift when nothing is planned passes on all 31 M5 versions, but only on 68 of 540 Dominick's and 38 of 362 Frat version-windows, with a median violation of 0.8% and 0.5% of the largest forecast. These failures come from two definitions of noth-

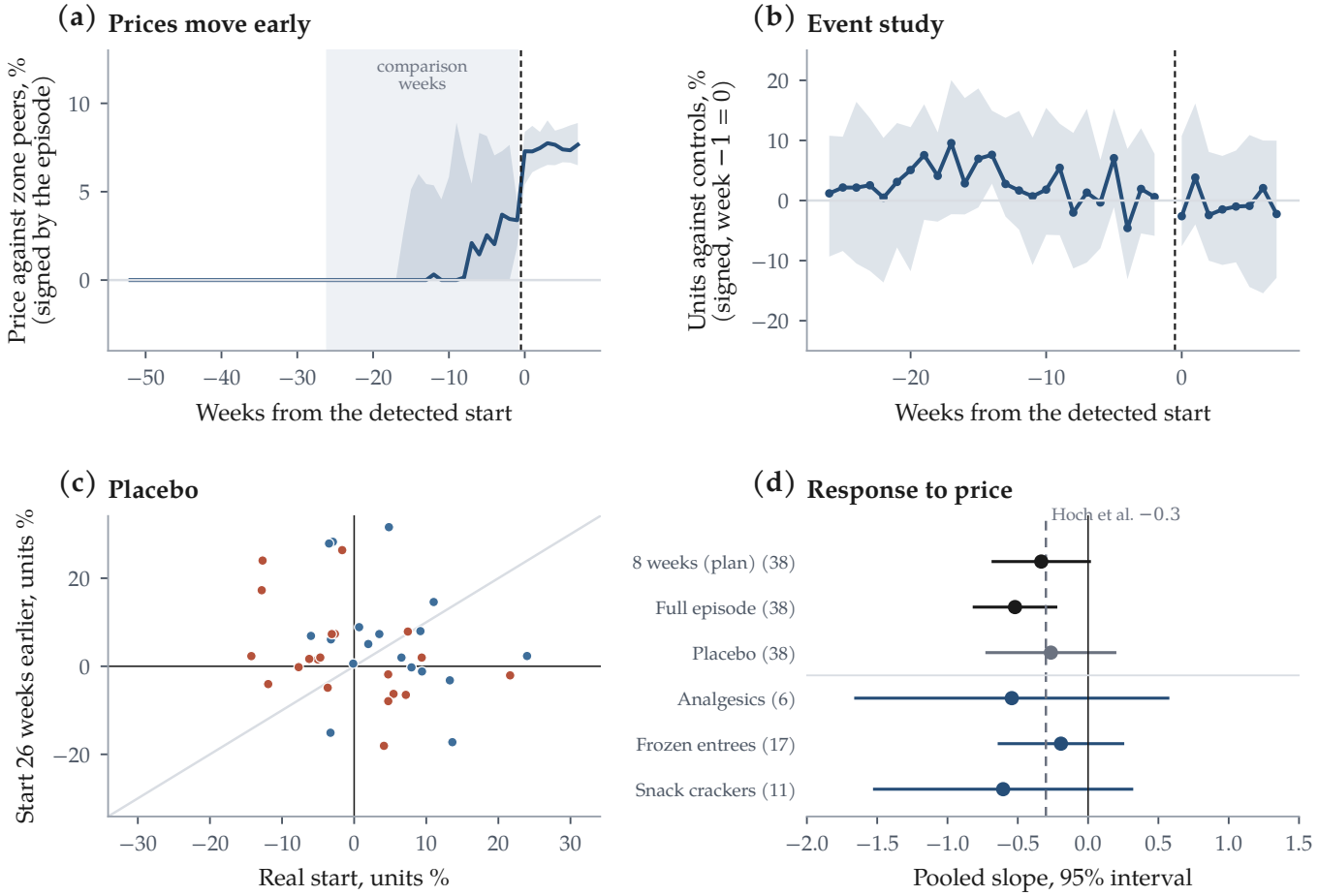



**Figure 8: Prices start moving before the detected start, single episodes are mostly noise, and only the pooled response carries information.** (**a**) The store's regular prices against its zone peers, signed by the episode's direction: median and interquartile band across the 38 scored episodes; shaded, the 26 comparison weeks of the difference in differences. (**b**) Event study: weekly coefficients signed the same way, week −1 as reference; median and interquartile band. (**c**) The 8-week estimate with the start moved 26 weeks earlier, where prices had not moved, against the real estimate; red, prices up; blue, down. (**d**) Pooled slopes of the response on the price change with 95% intervals, for the plan's 8-week estimate, the full episode, the placebo, and each category with at least five episodes (count in brackets). The detection rule, controls and estimator were fixed in advance; panels (a) and (d) were added after the placebo was seen.

**Table 4: No version passed every check on M5 or Dominick's.** Pass counts by check and data set, over version-windows (a check counted only where it applies: C3 needs two or more seeds, C6 a LightGBM run on the same window). Thresholds in Table 1.

| Data | Versions | Windows | C1 | C2 | C3 | C4 | C5 | C6 | All |
|---|---|---|---|---|---|---|---|---|---|
| M5 | 31 | 31 | 31/31 | 22/31 | 3/27 | 0/31 | 0/31 | 0/6 | 0/31 |
| Dominick's | 279 | 540 | 68/540 | 423/540 | 209/224 | 16/540 | 345/540 | 5/18 | 0/540 |
| Frat | 184 | 362 | 38/362 | 325/362 | 362/362 | 326/362 | 362/362 | 6/6 | 35/362 |

ing planned. The audit's plan turns the levers off and sets each price to the series' highest price over the last 12 training periods, keeping the planned price where the series has none. Each model has its own plan: the levers off and its own regular price, taken over its history window, from a recorded or a held regular price where its configuration says so, and otherwise the series' mean price over that window, or 1.0 where the window records no price. Given the model's own plan, the lift is zero for every series with history, because the plan path and the nothing-planned path then receive identical inputs; on saved Frat runs the violation fell from 0.04% to 0.09% of the largest forecast under the audit's plan to under $10^{-7}$ under the model's.

Whether the two plans differ depends on the data and the settings, not on the fitted weights, so we compared them for every audited configuration and window (933 of the 933 version-windows;

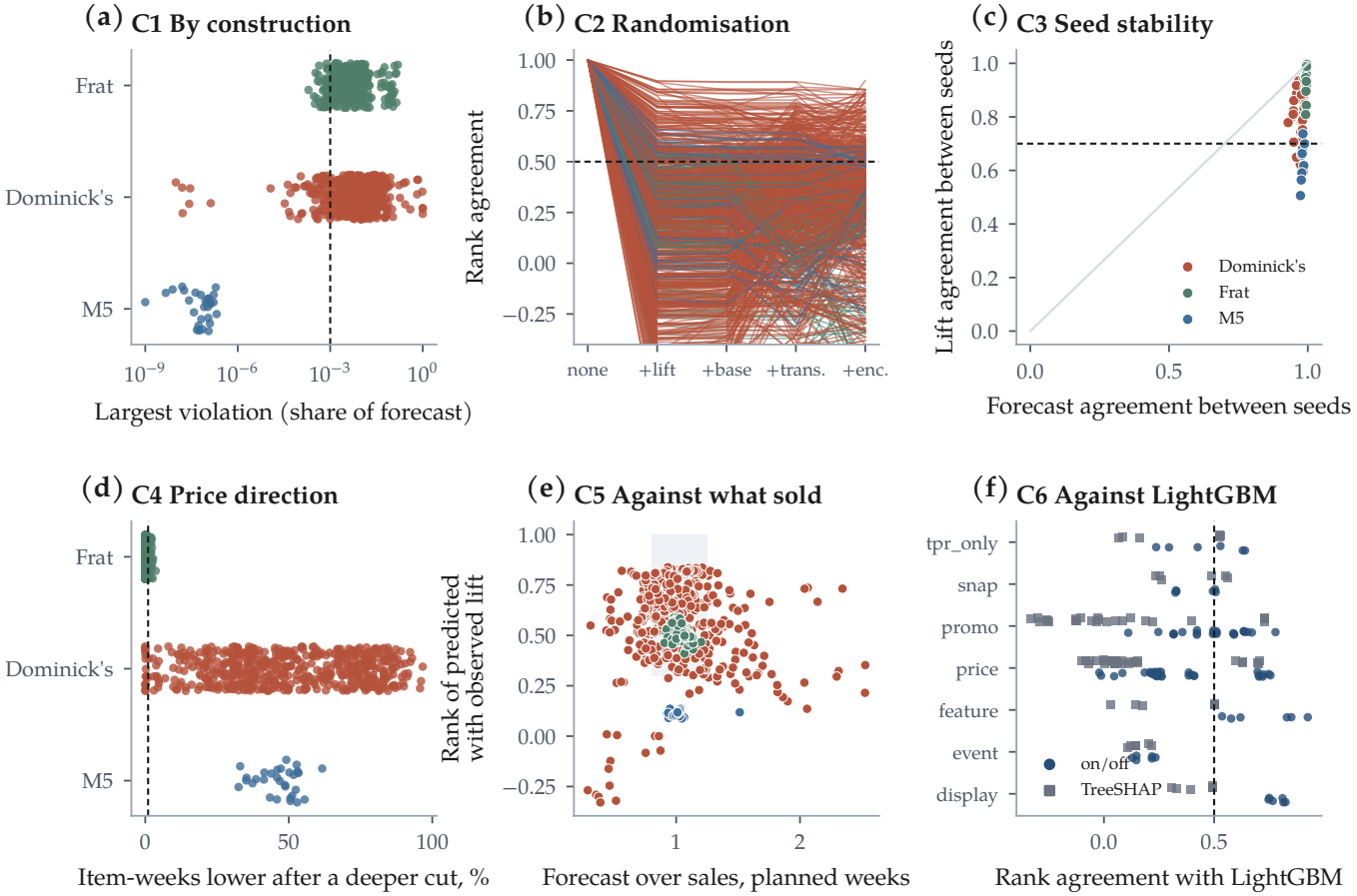



**Figure 9: The explanations agreed with themselves across seeds on Dominick's and Frat, failed the price-direction check on Dominick's and M5, and agreed with LightGBM only on Frat.** One point per version-window, coloured by data set; dashed lines: pass thresholds. (**a**) C1, the largest violation of the construction properties as a share of the largest forecast. (**b**) C2, rank agreement of the total lift with the original as the lift readout, baseline readout, transition and encoder are re-initialised in turn. (**c**) C3, agreement of lifts between seeds against agreement of forecasts. (**d**) C4, the share of item-weeks forecast lower after a deeper price cut. (**e**) C5, forecast over sales and rank of predicted with observed lift on planned item-weeks; shaded: the pass region. (**f**) C6, rank agreement with LightGBM's on/off contributions (circles) and TreeSHAP values (squares), by lever.

`scripts/c1_interface.py`, added after the results). They differ in 534 of 540 Dominick's, 362 of 362 Frat and 0 of 31 M5 version-windows (M5 checked on three of its ten stores, for memory), on a median of 0.1% of series-periods on Dominick's and 0.2% on Frat. Every one of the 796 version-windows that fail C1 has a difference; all 37 in which the plans agree pass, with violations no larger than 2e-7 of the largest forecast; 100 with a difference still pass, where the series affected carry little lift. Of the series-periods that differ, 37% are series with no price in the audit's last 12 periods, where the model uses its fallback (a price of 1.0 in 34% of all differing series-periods); 61% are configurations whose regular price comes from a held or a recorded price; and 2% are new items, whose baseline borrows several other series' regular prices, so that no single plan matches it. Where the two prices differ they are a median 2.7 times apart.

C1 under the audit's plan therefore measures the interface between the audit and the model, not a broken decomposition. It also shows a fault a merchant would care about: for an item with no recent price, the model's baseline is computed at a price nobody would plan, and the lift absorbs the difference. The audit's C1 scores and pass counts are reported unchanged in Table 4 and the main test.

**C2 and C3: the lifts depend on what was learned, and agree across seeds where the data are rich.** After the lift readout is re-initialised the lifts' rank agreement with the original falls below 0.5 in 423 of 540 Dominick's, 325 of 362 Frat and 22 of 31 M5 version-windows (medians 0.19, 0.29 and 0.29), so the lifts mostly depend on what the readout learned; in the known-answer study the same check did not separate correct models from broken ones (§4), so a pass here says little. Seeds agree on the lifts with a median rank agreement of 0.96 on Frat and 0.85 on Dominick's, but only 0.66 on M5, against 0.98 to 0.99 for the forecasts themselves on every data set. On M5, where prices rarely change and the plan is

mostly event and SNAP days, fits that agree on the forecast disagree on how much of it the plan explains; 3 of 27 M5 version-windows pass C3.

**C4: on Dominick's and M5 a deeper price cut often lowers the forecast.** A deeper cut gives a forecast lower by more than 0.1% in a median of 49% of Dominick's and 49% of M5 item-weeks, against 0% on Frat; 16 of 540 Dominick's and 0 of 31 M5 version-windows pass. The diagnosis rules out the two simple explanations. Cutting only a random tenth of series, so that other items' cuts barely enter, gives the same share (median 73% against 73% on Dominick's, 45% against 46% on M5). And the drops are not small: in a median of 57% of Dominick's and 11% of M5 item-weeks the forecast falls by more than 5%, and the share is already 71% and 34% at the first 5% cut. On Frat 0% of item-weeks fall by more than 5%. On Dominick's and M5 these models' forecast fell after a deeper cut in about half the item-weeks or more, and no accuracy score showed it. (The rerun runs, chosen partly for large C1 violations, show higher shares than the full audit on Dominick's.)

**C5 and C6: agreement with sales and with LightGBM.** On planned item-weeks the forecast is close to sales in total on every data set (median bias 0.98, 1.03 and 1.00), and the rank agreement of predicted with observed lift is 0.64 on Dominick's and 0.52 on Frat but 0.11 on M5 (event days 0.06, SNAP days 0.12), so no M5 version passes C5. Agreement with LightGBM's total lift passes on Frat (median 0.73) and mostly fails on Dominick's (0.39) and M5 (0.40). Lever by lever, the models' promotion contributions on Dominick's agree with LightGBM's on/off contributions (median rank agreement 0.50) but not with its TreeSHAP values ($-0.11$); on Frat they agree with both (0.50 and 0.46). Which attribution of LightGBM one compares with can change the verdict.

**C7: both models state a larger price response than the one measured.** Over the 38 scored zone price episodes, the pooled slope of the stated response on the price deviation is $-0.62$ for the world model (95% interval $-0.80$ to $-0.44$) and $-0.75$ for LightGBM ($-0.92$ to $-0.58$), against $-0.33$ ($-0.68$ to 0.01) measured over the same eight weeks and about $-0.3$ in the chain's own experiments (Figure 10). Both models state roughly twice the measured response, though the measured interval is wide and overlaps both, the pattern the known-answer study produced when a model answers a question about the regular price with what it learned from temporary cuts (§4). The measured interval is wide and overlaps the world model's, so these episodes do not exclude the world model's slope; both models' intervals lie below the $-0.3$ the experiments implied. Per episode, the world model's stated response falls inside the measured interval in 22 of 38 episodes and LightGBM's in 13, but those intervals are too narrow to trust (§6); the stated response has the expected sign in 32 and 31 episodes, the measured one in 25.

**The main test: on these versions, passing more checks did not clearly go with forecasting better.** On the unit fixed before any result, a version and window, the correlation across version-windows between the number of checks passed and item-by-store WRMSSE is $-0.32$ on M5, $-0.19$ on Dominick's and 0.26 on Frat; pooled, $-0.08$ (permutation $p = 0.205$; Table 5). Averaged over each version's windows the pooled correlation is $-0.19$ ($p = 0.004$). The answer moves with the unit, and neither permutation keeps the dependence in these data: versions that share a configuration, configurations that differ in one setting, and versions scored on the same window are not independent, and a window that is easier and passes C1 and C4 less often (the second Frat window) enters the version-window correlation as a relation between checks and error.

We therefore reran the test, as an analysis added after the results, on comparisons that hold the data and the window fixed. Seeds of a configuration are averaged, so a configuration counts once per window. Configurations scored on the same series and windows form a set (16 sets with at least four configurations, 147 configurations in all); within each set and window the Spearman correlation across configurations is taken, and the correlations are pooled weighted by the number of configurations. The permutation reassigns whole configurations, with their scores in every window, within each set, and the interval comes from resampling configurations within each set. No M5 set has four configurations, so M5 drops out. The pooled correlation is $-0.10$ (95% interval $-0.21$ to 0.02, $p = 0.10$): $-0.16$ on Dominick's and 0.13 on Frat. With C5 left out it is $-0.02$ ($-0.13$ to 0.12, $p = 0.80$), and with C1 and C5 left out $-0.12$ ($-0.24$ to 0.03, $p = 0.10$). C5 compares the forecast with sales over the same baseline and so shares information with accuracy; on the unit fixed in advance it is the check most strongly related to accuracy (rank of predicted with observed lift $-0.42$, bias 0.45).

What these data support is narrow. On these saved versions, configurations compared on the same data and window that passed more checks did not clearly forecast better or worse. The intervals run from about $-0.25$ to $0.12$: they exclude a strong relation in either direction, not a weak one. They do not show that faithfulness and accuracy are unrelated in general: the versions are development variants of two model families, the checks measure faithfulness only in part, and a stronger rela-

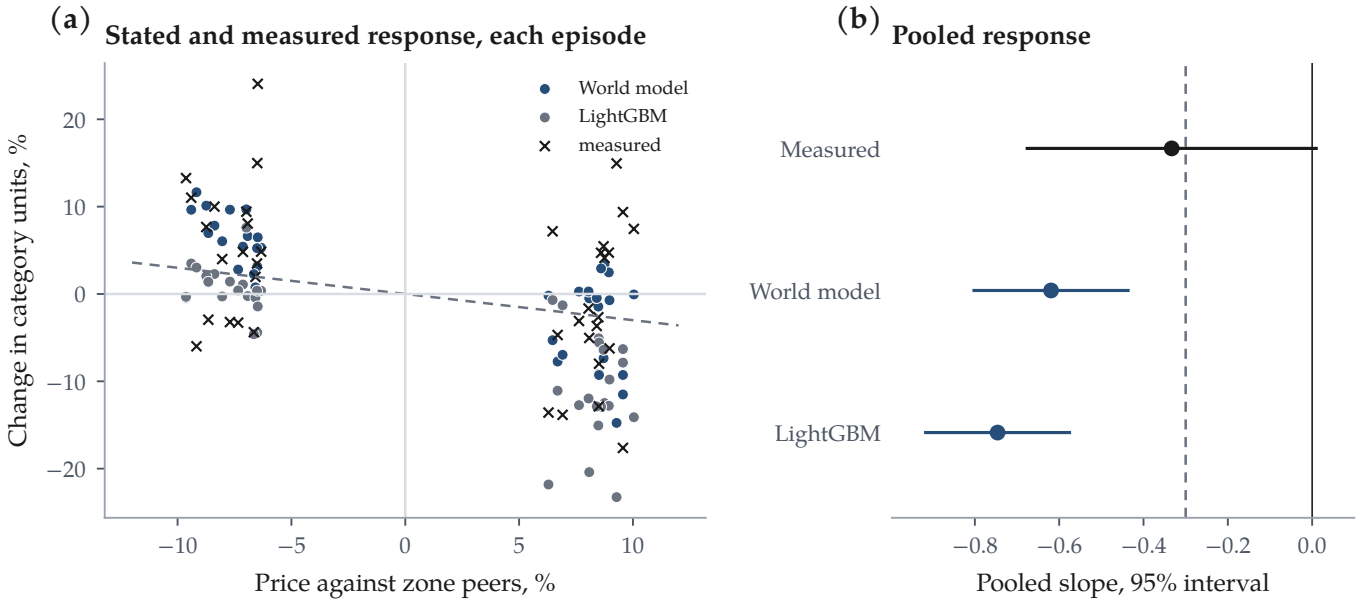



**Figure 10: The world model and LightGBM state about twice the measured response to a regular price change; the measured interval is wide.** (**a**) For each scored Dominick's zone price episode, the store's price against its zone peers and the change in its category units over the first eight weeks: stated by the world model and LightGBM, each fitted up to the week before the episode, and measured by difference in differences; dashed: an elasticity of −0.3. (**b**) Pooled slope across episodes with a 95% interval robust to unequal variances.

tion could hold across more different models.

## 8 Discussion

**The explanation was faithful but wrong in size and in kind.** The world model's explanation is faithful by construction: the baseline does not move with the plan, the lift is zero when nothing is planned, and the two add up to the forecast. In the known-answer study that explanation was right about what each lever adds when plans were made at random, and wrong in two ways that no check against the model could see. When merchants plan into strong weeks, the lifts are a third too large, because the data tie the plan to demand the model cannot see. And asked about a change in the regular price, the model answers with the response to a temporary cut, about five to six and a half times the truth. Molecular attributions in MolSanity [2] came apart from the truth for a similar reason: a model can be faithful to what it learned and what it learned can be wrong.

**What this means for a merchant reading a lift.** In the known-answer study, a lift read from a model fitted to the store's own history inherited its planning habits. If promotions go into weeks that would sell well anyway, the model's promotion lift overstates what the promotion adds, and an accurate forecast is no protection: in the known-answer study the correct world-model seeds and LightGBM forecast better when plans followed demand than when they were made at random. The check against what sold, the one most often run in practice, agrees with the overstated lift for the same reason. Two things help. Agreement between models fitted to the same data should not be read as confirmation. And a model's answer to a question it was not trained on, here a lasting change in the regular price from a model that mostly saw temporary cuts, should be checked against a measured response before it is used to set prices.

**Which checks to run.** The known-answer study ranks the checks. The price-direction check (C4) is cheap and caught every model blind to price, though it also flags LightGBM, whose response to price comes in steps. Per-lever agreement with a second model, an exploratory comparison, caught lifts given to the wrong lever, which every check on the total lift missed. Construction (C1) and seed stability (C3) are worth confirming once. The check against what sold (C5) and agreement with LightGBM on the total lift (C6) confirm agreement, not correctness. Randomisation (C2), the standard sanity check for attributions [8], does not test an explanation built into the model: re-initialising every part left the lifts' ranking largely in place. A version on the lift as a share of the baseline, or within planned weeks only, may work; we did not test it. A measured cause (C7) is the only check that found the regular-price error, though with six spells it could not tell a missing price response from a correct one. We expect that a measured promotion effect would be needed to find the bias from planning into strong weeks; we did not test this, and public

**Table 5: Across versions, the number of checks passed was not clearly related to accuracy; the check that was, C5, shares information with accuracy.** Spearman correlation between each score and item-by-store WRMSSE (negative: higher scores go with lower error), within each data set and pooled by permutation within data set (10,000 permutations). Left: over version-windows, the unit fixed in advance. Right: over versions, the mean over each version's windows. Both permutations treat their units as independent; the test that holds the data and window fixed and keeps the dependence between related configurations is in the text. †Added after the results.

| | Version and window | | | | | Version | |
|---|---|---|---|---|---|---|---|
| Score | M5 | Dom. | Frat | Pooled | $p$ | Pooled | $p$ |
| Checks passed | −0.32 | −0.19 | 0.26 | −0.08 | 0.205 | −0.19 | 0.004 |
| C1 violation | 0.10 | −0.18 | −0.41 | −0.17 | 0.007 | −0.03 | 0.664 |
| C2 agreement after randomising | −0.11 | 0.20 | −0.02 | 0.02 | 0.740 | 0.01 | 0.842 |
| C3 seed agreement | −0.07 | 0.32 | 0.30 | 0.18 | 0.010 | −0.01 | 0.938 |
| C4 share lower | −0.06 | −0.04 | −0.18 | −0.09 | 0.159 | −0.01 | 0.916 |
| C5 rank | −0.19 | −0.32 | −0.75 | −0.42 | 0.000 | −0.32 | 0.000 |
| C5 bias | 0.62 | 0.03 | 0.70 | 0.45 | 0.000 | 0.36 | 0.000 |
| C6 agreement with LightGBM | −0.31 | −0.48 | 0.20 | −0.20 | 0.394 | −0.20 | 0.392 |
| Checks passed, C5 left out† | −0.32 | −0.04 | 0.26 | −0.03 | 0.615 | −0.13 | 0.055 |
| Checks passed, C1 and C5 left out† | −0.32 | −0.10 | 0.15 | −0.09 | 0.163 | −0.15 | 0.032 |

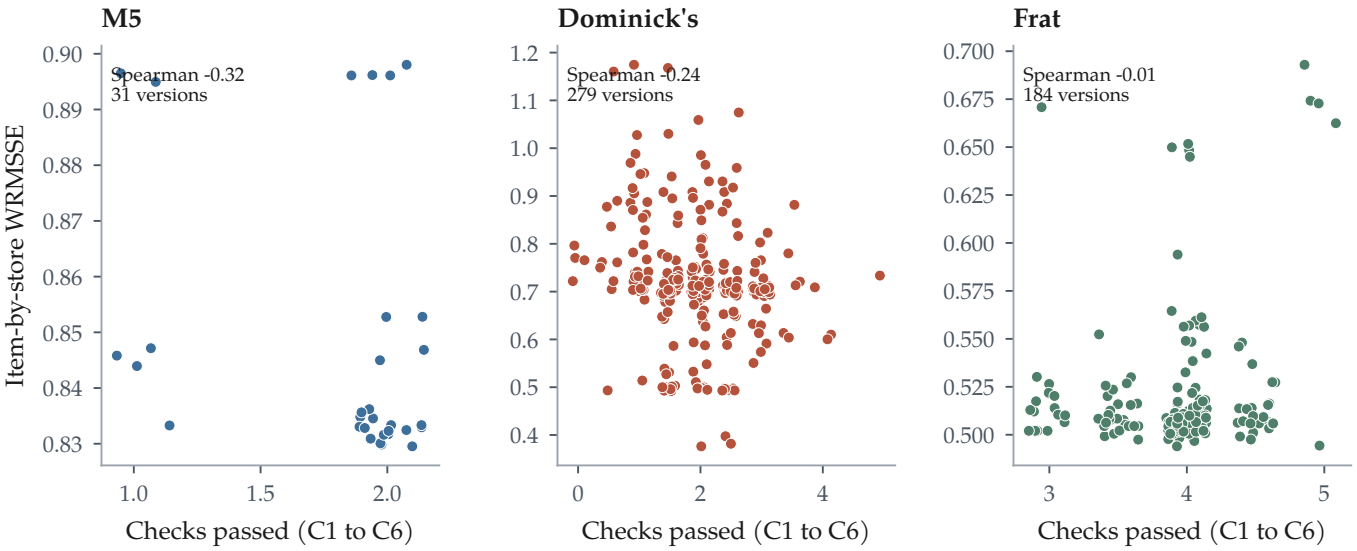



**Figure 11: Checks passed against accuracy, one point per version.** Item-by-store WRMSSE (lower is better) against the mean number of checks passed over the version's windows, with the Spearman correlation within each data set.

retail data offer none.

**Measured causes on public data are scarce and noisy.** Dominick's zone price deviations give a pooled 8-week response close to a placebo at false dates and a full-spell response in line with the chain's own experiments; single episodes are noise, intervals clustered by a handful of stores are far too narrow, and prices start moving before the detector sees them. A retailer running its own tests would do better with the design under its control: stores assigned at random, a known start date, and enough stores per arm for honest intervals.

**The audit of the saved versions.** The real versions behaved like the known-answer models in the two ways that matter most. Checks against the model and against another model were not shown to track whether the lift is right, and the one comparison with a measured cause found both models stating a larger price response than the measured one, as the known-answer world model did, though the measured interval is wide. The real versions also showed two things the generated data did not. First, the lift was not zero when the audit planned nothing on most Dominick's and Frat versions. Every such violation traces to series and periods where the audit's plan differs from the model's own; under the model's own plan the lift is zero. Part of the difference is the

regular-price rule a configuration chose; the rest is a fault in the baseline: for an item with no recent price the model computes its baseline at the item's mean price, or at a price of 1.0, which no merchant would call nothing planned. An audit has to read the model's own definition of nothing planned, and check it, before C1 means anything. Second, the price-direction check failed on most Dominick's and M5 versions because their forecasts fall after a deeper cut in about half the item-weeks or more, a fault no accuracy score revealed and one a merchant setting prices from the model would meet directly. Across versions, passing more checks did not clearly go with forecasting better. For a retailer this means two separate tests: an accuracy comparison to choose the forecast, and checks of the explanation, with a price-direction test and a measured cause among them, before its lifts are used to plan.

## 9 Conclusion

Plan-lift explanations built into a forecasting model can be faithful, stable and in agreement with sales and with another model, and still be wrong about what a plan causes. In generated store data with known effects, the checks that compare an explanation with the model, the data or another model caught a model blind to price and, compared lever by lever, lifts given to the wrong lever. They did not catch lifts inflated by planning into strong weeks, or a model that answers a question about the regular price with the response to a temporary cut. Only a measured cause found the second. On public retail data the one such cause we could measure, Dominick's zone price deviations, gives a pooled response near the chain's experiments over the full spell, but imprecise. In the audit of 494 saved versions on public data, no version passed every check on M5 or Dominick's, both the world model and LightGBM stated about twice the measured response to a regular price change (the measured interval is wide and overlaps both), and, among configurations compared on the same data and window, the number of checks passed was not clearly related to how well they forecast. On these models and data sets, an accurate forecast was not evidence of a correct explanation, and the explanation has to be tested on its own.

## 10 Limitations and threats to validity

**The benchmark is observational and covers one effect on one data set.** C7 compares a model with a response estimated from observational data: the treated stores and weeks are inferred, and the 8-week estimate is close to a placebo at false dates. The causal failure C7 is meant to catch is shown in the known-answer study, where the truth is known; on Dominick's, C7 can only tell a response several times too large from a plausible one. It measures the response of a store's category units to a sustained change in its regular prices. It says nothing directly about promotions, displays or temporary cuts, or about M5 and Frat, where no such variation is used. A model can be right about regular prices and wrong about promotions; the known-answer study shows how far the other checks go in covering the rest.

**The deviations are inferred.** The public files do not mark the experimental stores or weeks. Some deviations may be zone moves made for other reasons, and a store moved for a reason that also moves its sales would bias $\tau_e$. The event study and the placebo address this for trends that start before the price moves; they cannot rule out a shock that starts in the same week.

**Few control stores.** Zones have few stores, so each scored episode has 3 to 24 controls. Intervals clustered by store with so few clusters are wide and only approximately calibrated; the pooled slope across episodes carries the comparison.

**The baseline is a choice.** Every lift is measured against nothing planned, with the regular price taken as the highest of the last 12 periods. A different rule gives different baselines and lifts. The rule is the models' own and is applied the same way to every model and to the truth in the known-answer study.

**The versions are not a designed sample.** The audited versions are what the companion paper saved. They share data and differ in many settings at once, so the correlation across versions is descriptive; it is computed within data set and tested by permutation within data set, but versions are not independent draws, and the windows of one version differ in difficulty, so the result depends on whether version-windows or versions are the unit. Both are reported.

**The audit's definition of nothing planned.** The audit builds "nothing planned" from the data: levers off, the price at the series' highest price over the last 12 training periods, and the planned price where the series has no price in those periods. The models build their own from their history window and fall back to the series' mean price over that window, or to 1.0 where it records no price. C1 under the audit's plan therefore measures the distance between two definitions as much as the model; under the model's own plan it holds by construction (§7). Neither definition is the retailer's regular price, which these data do not record.

**Refitted versions.** The Dominick's final world models are refits with unchanged settings, not the runs whose scores the companion paper reports; seed-to-seed differences between the two are expected and are measured by C3.

**The generated world is simpler than retail.** The known-answer data are multiplicative in the plan, with one shock process and no cannibalisation between items. They test whether a check can catch a failure, not how often the failure occurs in practice.

## 11 Reproducibility, compute and licence

Every number, table and figure in this paper is generated from committed records by `make -C paper2`, and a continuous integration job rebuilds the paper from a fresh checkout on every change and fails on any LaTeX error, undefined reference or missing font. The checks are in `src/rwm/audit/`; the zone detector and estimators in `src/rwm/audit/zones.py`; the generated data in `src/rwm/data/synthetic_levers.py`. The runs:

`scripts/zone_episodes.py` lists the episodes and their responses (`results/paper2/zone_episodes.csv`);

`scripts/zone_robustness.py` the event study, placebo and sensitivity;

`scripts/known_answer.py` the known-answer study;

`scripts/audit_runs.py` and `scripts/paper2_colab.sh`: the audit of the saved versions, C6 and C7;

`scripts/audit_diagnose.py` the diagnosis of C1 and C4 (`results/paper2/diagnose/`);

`scripts/c1_interface.py` where the audit's plan for nothing planned and each model's own differ, for every audited configuration and window (`results/paper2/diagnose/c1_interface.csv`);

`scripts/audit_summary.py` the tables committed under `results/paper2/`.

**Compute.** The known-answer study fits 14 models on a two-core processor: each world model in 5 to 23 minutes (69 minutes in all, the longest while another job shared the processor), each LightGBM in seconds. The zone episode list, event study, placebo, sensitivity and price path read only the data and take minutes on the same processor. The audit of the saved versions runs on a Colab GPU; its run times are recorded with its results.

**Copyright and licence.** Copyright © 2026 Karthik Iyer. Code under the PolyForm Noncommercial License 1.0.0; text, figures and tables under CC BY-NC 4.0. Industry and commercial use of the code, the manuscript or any part of them, including use inside a company, is prohibited unless approved in writing by the author.

**Data availability.** M5 is available from Kaggle [24], Dominick's from the Kilts Center for Marketing at the University of Chicago Booth School of Business [22], and Breakfast at the Frat from dunnhumby [23], each under its own terms; the repository fetches them and redistributes none.

## References

[1] A. Jacovi and Y. Goldberg. Towards faithfully interpretable NLP systems: How should we define and evaluate faithfulness? *Proceedings of the 58th Annual Meeting of the ACL*, pages 4198–4205, 2020.

[2] K. Iyer. MolSanity: a reliability audit for feature attributions on molecular graph neural networks. https://github.com/Kar488/molsanity.

[3] S. J. Hoch, X. Drèze, and M. E. Purk. EDLP, Hi-Lo, and margin arithmetic. *Journal of Marketing*, 58(4):16–27, 1994.

[4] M. T. Ribeiro, S. Singh, and C. Guestrin. "Why should I trust you?" Explaining the predictions of any classifier. *Proceedings of the 22nd ACM SIGKDD Conference*, pages 1135–1144, 2016.

[5] S. M. Lundberg and S.-I. Lee. A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems 30*, 2017.

[6] S. M. Lundberg, G. Erion, H. Chen, A. DeGrave, J. M. Prutkin, B. Nair, R. Katz, J. Himmelfarb, N. Bansal, and S.-I. Lee. From local explanations to global understanding with explainable AI for trees. *Nature Machine Intelligence*, 2(1):56–67, 2020.

[7] A. Adadi and M. Berrada. Peeking inside the black-box: A survey on explainable artificial intelligence (XAI). *IEEE Access*, 6:52138–52160, 2018.

[8] J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt, and B. Kim. Sanity checks for saliency maps. *Advances in Neural Information Processing Systems 31*, 2018.

[9] P.-J. Kindermans, S. Hooker, J. Adebayo, M. Alber, K. T. Schütt, S. Dähne, D. Erhan, and B. Kim. The (un)reliability of saliency methods. In *Explainable AI: Interpreting, Explaining and Visualizing Deep Learning*, LNCS 11700, pages 267–280. Springer, 2019.

[10] D. Alvarez-Melis and T. S. Jaakkola. On the robustness of interpretability methods. arXiv:1806.08049, 2018.

[11] S. Hooker, D. Erhan, P.-J. Kindermans, and B. Kim. A benchmark for interpretability methods in deep neural networks. *Advances in Neural Information Processing Systems 32*, 2019.

[12] C. Rudin. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence*, 1(5):206–215, 2019.

[13] T. Rojat, R. Puget, D. Filliat, J. Del Ser, R. Gelin, and N. Díaz-Rodríguez. Explainable artificial intelligence (XAI) on time series data: A survey. arXiv:2104.00950, 2021.

[14] A. A. Ismail, M. Gunady, H. Corrada Bravo, and S. Feizi. Benchmarking deep learning interpretability in time series predictions. *Advances in Neural Information Processing Systems 33*, pages 6441–6452, 2020.

[15] S. Tonekaboni, S. Joshi, K. Campbell, D. Duvenaud, and A. Goldenberg. What went wrong and when? Instance-wise feature importance for time-series black-box models. *Advances in Neural Information Processing Systems 33*, pages 799–809, 2020.

[16] A. L. Montgomery. Creating micro-marketing pricing strategies using supermarket scanner data. *Marketing Science*, 16(4):315–337, 1997.

[17] E. Abrams, G. Gui, and A. Hortaçsu. Finding exogenous variation in data. arXiv:1704.07787, 2017.

[18] J. D. Angrist and J.-S. Pischke. *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press, 2009.

[19] M. Bertrand, E. Duflo, and S. Mullainathan. How much should we trust differences-in-differences estimates? *Quarterly Journal of Economics*, 119(1):249–275, 2004.

[20] A. C. Cameron and D. L. Miller. A practitioner's guide to cluster-robust inference. *Journal of Human Resources*, 50(2):317–372, 2015.

[21] S. J. Hoch, B.-D. Kim, A. L. Montgomery, and P. E. Rossi. Determinants of store-level price elasticity. *Journal of Marketing Research*, 32(1):17–29, 1995.

[22] James M. Kilts Center for Marketing. Dominick's dataset. University of Chicago Booth School of Business. https://www.chicagobooth.edu/research/kilts/research-data/dominicks.

[23] dunnhumby. Breakfast at the Frat. dunnhumby source files. https://www.dunnhumby.com/source-files/.

[24] S. Makridakis, E. Spiliotis, and V. Assimakopoulos. M5 accuracy competition: Results, findings, and conclusions. *International Journal of Forecasting*, 38(4):1346–1364, 2022.

[25] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu. LightGBM: A highly efficient gradient boosting decision tree. *Advances in Neural Information Processing Systems 30*, 2017.

[26] B. Lim, S. Ö. Arık, N. Loeff, and T. Pfister. Temporal fusion transformers for interpretable multi-horizon time series forecasting. *International Journal of Forecasting*, 37(4):1748–1764, 2021.

[27] R. Fildes, S. Ma, and S. Kolassa. Retail forecasting: Research and practice. *International Journal of Forecasting*, 38(4):1283–1318, 2022.

[28] L. G. Cooper, P. Baron, W. Levy, M. Swisher, and P. Gogos. PromoCast: A new forecasting method for promotion planning. *Marketing Science*, 18(3):301–316, 1999.

[29] H. J. van Heerde, P. S. H. Leeflang, and D. R. Wittink. The estimation of pre- and postpromotion dips with store-level scanner data. *Journal of Marketing Research*, 37(3):383–395, 2000.

[30] S. Macé and S. A. Neslin. The determinants of pre- and postpromotion dips in sales of frequently purchased goods. *Journal of Marketing Research*, 41(3):339–350, 2004.

[31] S. Ma, R. Fildes, and T. Huang. Demand forecasting with high dimensional data: The case of SKU retail sales forecasting with intra- and inter-category promotional information. *European Journal of Operational Research*, 249(1):245–257, 2016.

[32] S. Gupta. Impact of sales promotions on when, what, and how much to buy. *Journal of Marketing Research*, 25(4):342–355, 1988.

[33] H. J. van Heerde, S. Gupta, and D. R. Wittink. Is 75% of the sales promotion bump due to brand switching? No, only 33% is. *Journal of Marketing Research*, 40(4):481–491, 2003.

[34] K. Iyer and M. Dabbeeru. World models, trees and foundation models for retail forecasting: evidence from four public data sets. Companion paper, https://github.com/Kar488/retail-world-models, 2026.

[35] K. Iyer. Paper 2 plan, written before any audit result, with dated amendments. `docs/PAPER2_PLAN.md` in https://github.com/Kar488/retail-world-models, 2026.

[36] H. Chen, J. D. Janizek, S. Lundberg, and S.-I. Lee. True to the model or true to the data? arXiv:2006.16234, 2020.

[37] D. Janzing, L. Minorics, and P. Blöbaum. Feature relevance quantification in explainable AI: A causal problem. *Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics*, PMLR 108:2907–2916, 2020.

[38] Z. C. Lipton. The mythos of model interpretability. *Communications of the ACM*, 61(10):36–43, 2018.

[39] F. Doshi-Velez and B. Kim. Towards a rigorous science of interpretable machine learning. arXiv:1702.08608, 2017.

[40] T. H. A. Bijmolt, H. J. van Heerde, and R. G. M. Pieters. New empirical generalizations on the determinants of price elasticity. *Journal of Marketing Research*, 42(2):141–156, 2005.

## A Every zone price episode

**Table 6: All 54 episodes the detector finds, with their measured response.** Weeks: Dominick's week numbers of the episode. Price: the store's median log price against its zone peers over the episode, %. Controls: same-zone stores within 1% of their zone. 8 weeks: difference-in-differences estimate on log category units over the first eight weeks, %, with its interval clustered by store (too narrow, §6); Full: over the whole episode. * Fewer than three controls: listed, not scored.

| Category | Zone | Store | Weeks | Price | Controls | 8 weeks | 95% interval | Full |
|---|---|---|---|---|---|---|---|---|
| Analgesics | 2 | 50 | 190–215 | 8.8 | 16 | 4.1 | 0.6 to 7.6 | −5.7 |
| Analgesics | 2 | 49 | 197–209 | 8.5 | 16 | −2.7 | −5.6 to 0.3 | −3.2 |
| Analgesics | 2 | 48 | 203–218 | 8.1 | 11 | −5.0 | −9.8 to −0.2 | −7.1 |
| Analgesics | 7 | 109 | 203–218 | 6.0 | 2 | 8.2 | −4.1 to 20.4 | 9.1 * |
| Analgesics | 1 | 32 | 204–218 | −6.5 | 7 | 13.6 | 6.7 to 20.6 | 14.9 |
| Analgesics | 2 | 88 | 204–218 | −6.7 | 10 | −3.5 | −9.1 to 2.2 | −2.0 |
| Analgesics | 2 | 89 | 204–218 | 6.3 | 10 | −12.7 | −18.3 to −7.1 | −9.4 |
| Bottled juice | 12 | 105 | 9–24 | −7.9 | 7 | −7.2 | −18.9 to 4.4 | −3.2 * |
| Bottled juice | 1 | 123 | 14–30 | −5.5 | 11 | 7.8 | 3.6 to 12.1 | 11.5 * |
| Bottled juice | 6 | 40 | 17–28 | −6.6 | 6 | 8.2 | −9.0 to 25.3 | 5.8 * |
| Bottled juice | 6 | 40 | 63–75 | −7.7 | 6 | −3.2 | −13.5 to 7.1 | −0.1 |
| Bottled juice | 2 | 84 | 266–283 | −9.4 | 24 | 11.0 | 8.9 to 13.2 | 12.7 |
| Bottled juice | 2 | 84 | 296–315 | −9.2 | 22 | −6.0 | −12.3 to 0.3 | −0.9 |
| Cheese | 6 | 77 | 249–261 | −7.1 | 7 | 4.8 | −12.3 to 21.9 | 4.1 |
| Frozen entrees | 5 | 8 | 201–218 | 10.2 | 1 | −26.8 | −26.8 to −26.8 | −31.0 * |
| Frozen entrees | 2 | 47 | 202–218 | 6.7 | 9 | −4.7 | −10.1 to 0.7 | −3.2 |
| Frozen entrees | 2 | 28 | 204–218 | −6.9 | 9 | 8.0 | 2.4 to 13.5 | 7.9 |
| Frozen entrees | 1 | 71 | 204–218 | 9.3 | 4 | 21.6 | 12.7 to 30.6 | 17.1 |
| Frozen entrees | 2 | 91 | 204–218 | −8.4 | 9 | 6.6 | 1.0 to 12.1 | 9.7 |
| Frozen entrees | 6 | 40 | 205–218 | −7.6 | 2 | −3.2 | −6.8 to 0.4 | −2.1 * |
| Frozen entrees | 12 | 114 | 205–218 | −8.7 | 3 | 9.2 | −5.7 to 24.0 | 7.5 |
| Frozen entrees | 2 | 119 | 205–218 | −8.0 | 9 | −0.1 | −5.6 to 5.4 | −1.1 |
| Frozen entrees | 1 | 2 | 220–237 | 9.0 | 4 | −6.2 | −11.0 to −1.5 | −12.8 |
| Frozen entrees | 5 | 8 | 220–237 | 9.8 | 2 | −13.0 | −95.9 to 69.9 | −7.7 * |
| Frozen entrees | 2 | 9 | 220–237 | −9.6 | 9 | 13.3 | 9.4 to 17.2 | 7.8 |
| Frozen entrees | 7 | 33 | 220–237 | 5.9 | 2 | 2.3 | 1.0 to 3.6 | −1.1 * |
| Frozen entrees | 6 | 40 | 220–237 | −10.9 | 1 | 21.0 | 21.0 to 21.0 | 15.6 * |
| Frozen entrees | 2 | 49 | 220–234 | 8.9 | 9 | 4.7 | 0.8 to 8.6 | 5.0 |
| Frozen entrees | 1 | 68 | 220–237 | 7.6 | 4 | −3.1 | −7.8 to 1.7 | −12.2 |
| Frozen entrees | 1 | 71 | 220–237 | 10.0 | 4 | 7.4 | 2.7 to 12.2 | −5.4 |
| Frozen entrees | 7 | 75 | 220–234 | −10.2 | 2 | −0.3 | −1.6 to 1.0 | −4.7 * |
| Frozen entrees | 2 | 76 | 220–234 | 8.6 | 9 | 4.7 | 0.8 to 8.6 | 4.8 |
| Frozen entrees | 1 | 95 | 220–234 | 8.1 | 4 | −1.7 | −6.4 to 3.1 | −6.4 |
| Frozen entrees | 2 | 110 | 220–234 | 8.4 | 9 | −3.7 | −7.6 to 0.2 | −1.6 |
| Frozen entrees | 2 | 119 | 220–237 | −6.6 | 9 | 2.0 | −1.9 to 5.9 | −2.8 |
| Frozen entrees | 1 | 130 | 220–233 | −7.3 | 4 | −3.3 | −8.0 to 1.5 | 2.1 |
| Frozen entrees | 1 | 71 | 270–290 | −7.0 | 14 | 9.4 | 6.0 to 12.8 | 5.0 |
| Soft drinks | 12 | 105 | 1–20 | −7.4 | 7 | 18.0 | 18.0 to 18.0 | 8.8 * |
| Snack crackers | 2 | 50 | 196–216 | 9.6 | 10 | 9.4 | 3.2 to 15.6 | −13.1 |
| Snack crackers | 2 | 48 | 197–218 | 8.5 | 9 | −12.8 | −18.5 to −7.1 | −13.2 |
| Snack crackers | 2 | 76 | 197–218 | 6.9 | 9 | −11.9 | −18.1 to −5.8 | −6.8 |
| Snack crackers | 2 | 49 | 198–218 | 9.6 | 9 | −14.3 | −19.2 to −9.3 | −6.6 |
| Snack crackers | 2 | 89 | 198–218 | 8.5 | 9 | −7.7 | −12.6 to −2.8 | −1.1 |
| Snack crackers | 2 | 47 | 200–218 | 6.5 | 9 | 7.2 | 2.7 to 11.6 | 9.6 |
| Snack crackers | 2 | 28 | 203–218 | −6.5 | 9 | 24.0 | 20.2 to 27.8 | 22.4 |
| Snack crackers | 5 | 94 | 203–218 | −7.1 | 2 | −1.2 | −8.0 to 5.7 | −6.2 * |
| Snack crackers | 1 | 2 | 204–218 | 8.7 | 4 | 5.5 | −7.4 to 18.3 | 2.9 |
| Snack crackers | 7 | 75 | 204–218 | −7.4 | 2 | −9.9 | −10.9 to −8.8 | −12.6 * |
| Snack crackers | 2 | 88 | 204–218 | −8.6 | 9 | −2.9 | −7.2 to 1.3 | −0.6 |
| Snack crackers | 12 | 114 | 204–218 | −6.7 | 2 | 9.2 | −2.7 to 21.0 | 6.0 * |
| Snack crackers | 2 | 119 | 204–218 | −6.5 | 9 | 3.5 | −0.8 to 7.7 | 4.1 |
| Snack crackers | 2 | 113 | 206–218 | −6.3 | 9 | 0.7 | −3.9 to 5.3 | 0.1 |
| Snack crackers | 6 | 40 | 207–218 | −8.7 | 2 | 9.5 | −18.8 to 37.8 | 14.6 * |
| Snack crackers | 12 | 129 | 207–218 | −7.4 | 2 | 2.1 | −4.9 to 9.0 | −0.9 * |

## B The plan and its amendments

The plan (`docs/PAPER2_PLAN.md`) was committed before any audit result on 13 October 2026. Four dated amendments followed, each made before the results it affects were read:

1. Before any model lift: the episode detector measures each store against its zone peers rather than the zone's most common price.
2. After the episode list, before any model lift: at least three controls to score an episode; models scored on an episode's first eight weeks, against an 8-week difference in differences.
3. Before any audit result: the known-answer study, the C7 robustness analyses, the pass thresholds and the main test.

4. After the C7 robustness results, before any model lift for C7: the price path and the pooled slope intervals, labelled exploratory, and the comparison of models with the measured response made on the pooled slope.

## C Every known-answer model

**Table 7: Every model in the known-answer study, with its explanation scored against the truth.** Lift against truth: rank agreement and ratio of summed lifts on item-weeks with something planned, overall, and the ratio for the price part. C4: item-weeks lower after a deeper cut. C5: summed forecast over summed sales on planned weeks. Passed: checks passed of those that apply (C1 to C6). RMSSE: item by store, weeks 253 to 260.

| | Lift against truth | | | | | | |
|---|---|---|---|---|---|---|---|
| Model | rank | sum | price | C4 | C5 | Passed | RMSSE |
| *Plans at random* | | | | | | | |
| World model, seed 1 | 0.99 | 1.14 | 1.13 | 0.0% | 1.08 | 6/6 | 0.401 |
| World model, seed 2 | 0.99 | 1.10 | 1.09 | 0.0% | 1.06 | 6/6 | 0.401 |
| World model, seed 3 | 0.99 | 1.11 | 1.11 | 0.0% | 1.05 | 5/6 | 0.398 |
| World model, blind to price | 0.59 | 0.73 | 0.01 | 31.5% | 0.90 | 3/5 | 0.510 |
| World model, blind to levers | 0.89 | 1.10 | 1.35 | 0.0% | 1.09 | 4/5 | 0.450 |
| LightGBM | 0.85 | 1.09 | 1.03 | 7.6% | 0.99 | 1/2 | 0.389 |
| LightGBM, blind to price | 0.43 | 0.73 | −0.00 | 49.5% | 0.88 | 1/2 | 0.515 |
| *Plans in high-demand weeks* | | | | | | | |
| World model, seed 1 | 0.96 | 1.33 | 1.22 | 0.0% | 1.06 | 5/6 | 0.345 |
| World model, seed 2 | 0.95 | 1.35 | 1.24 | 0.0% | 1.07 | 6/6 | 0.349 |
| World model, seed 3 | 0.95 | 1.40 | 1.27 | 0.0% | 1.09 | 5/6 | 0.344 |
| World model, blind to price | 0.51 | 1.09 | −0.00 | 27.2% | 0.93 | 2/5 | 0.489 |
| World model, blind to levers | 0.82 | 1.23 | 1.63 | 0.0% | 1.07 | 4/5 | 0.480 |
| LightGBM | 0.82 | 1.26 | 1.13 | 9.9% | 1.01 | 1/2 | 0.322 |
| LightGBM, blind to price | 0.54 | 0.94 | −0.02 | 43.4% | 0.90 | 1/2 | 0.497 |